

Experiment 1: FIND-S Algorithm**Aim**

To implement the FIND-S algorithm for finding the most specific hypothesis based on a given set of training data samples.

Algorithm

Step 1: Import the required libraries.

Step 2: Read the training dataset from the trainingdata.csv file.

Step 3: Initialize the hypothesis with the first positive training example.

Step 4: Traverse through all training examples.

Step 5: If the example is positive ("Yes"), compare each attribute with the current hypothesis.

- If the attributes are the same, retain the value.
- If they differ, replace the attribute with '?'.

Step 6: Ignore all negative examples.

Step 7: Display the final most specific hypothesis.

Program

```
#Experiment 1 - FIND-S Algorithm
import csv

with open('trainingdata.csv') as csvFile:
    data = list(csv.reader(csvFile))

hypothesis = data[1][:-1]

for row in data[1:]:
    if row[-1].lower() == "yes":
        for i in range(len(hypothesis)):
            if hypothesis[i] != row[i]:
                hypothesis[i] = '?'

print("Most Specific Hypothesis:")
print(hypothesis)
```

Output

```
... Most Specific Hypothesis:  
['Sunny', 'Warm', '?', 'Strong', '?', '?']
```

Result

Thus, the FIND-S algorithm was successfully implemented, and the most specific hypothesis consistent with the given positive training examples was obtained.